Marshall Bruner

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RF hardware, signal processing, and deployment | Teaching 50k+ engineers on YouTube

EXPERIENCE

Rainmaker Technology Corporation **Aug 2025 — Oct 2026**
Head Radar Systems Engineer *Boulder, CO*

- Led radar engineering team building in-house weather radars across S-, X-, and K-band systems
- Designed a 24 GHz FMCW cloud-profiling radar from RF architecture through PCB layout, fab, and bring-up
- Operated and deployed a 25 kW X-band dual-pol weather radar, including a mountain-top installation
- Built embedded control server, and signal processing pipeline
- Performed global radar analysis (incl. building internal tools) for BD team

CSU-CHILL National Radar Facility **Jan 2025 — Present**
Radar Systems Engineer (part-time) *Greeley, CO*

- Built the facility's real-time multi-radar web display, serving 8+ radars
- Wrote the antenna positioner controller for a solid-state C-band radar, from pedestal protocol to operator UI
- Relocated and re-commissioned a dual-pol S-band weather radar at Kwajalein Atoll, Marshall Islands

BAE Systems (DoD Clearance) **Jul 2024 — Jul 2025**
RF Engineer *Westminster, CO*

- Designed, laid out, and brought up RF boards for electronic warfare systems, from system design onward
- Designed a 2D active phased array weather radar, from system architecture to schematic and layout
- Analyzed large-scale phased array systems
- Consulted on weather radar proposals

Ball Aerospace (DoD Clearance) **Jun 2021 — Jul 2023**
RF Engineer *Westminster, CO*

- Designed and debugged multiple mission-critical RF boards under strict time constraints
- Simulated, designed, and tested an RF front-end PCB at extreme temperatures
- Schematic capture and layout of RF T/R systems, antenna range test equipment, and test coupons
- Built an embedded webpage to control and monitor phased array electronics

Colorado State University **Jan 2021 — May 2022**
Graduate Research Assistant | Thesis focused on Radar *Fort Collins, CO*

- Designed and laid out a modular X-band FMCW drone radar
- Signal processing and visualization for Analog Devices' FMCW Phaser board, shown by ADI at IMS 2022
- Presented at the IEEE International Symposium on Phased Array Systems and Technology

CSU-CHILL Radar Lab **May 2020 — May 2021**
Lab Assistant *Greeley, CO*

- Built a server to host and display real-time and historical radar data
- Characterized and measured radar equipment

PROJECTS

Marshall Bruner RF — Radar / RF Education YouTube Channel **Aug 2023 — Present**

- 53k subscribers and 1.4M+ views across 17 animated videos on radar, phased arrays, and signal processing
- Write, animate, narrate, and edit every video; built a reusable radar animation library
- Ship companion interactive notebooks and a 3D weather-radar animation tool

FMCW Radar Bootcamp — IEEE AESS RadarConf **Mar 2023**

- Developed signal processing software and a presentation for the 2023 AESS Radar Bootcamp

EDUCATION

Colorado State University **Jan 2021 — May 2022**
M.S. Electrical Engineering — ***GPA: 3.9*** *Fort Collins, CO*

Thesis - Design, Deployment, and Cost Considerations for DARMA; A Low-Cost and Lightweight FMCW Radar

Colorado State University **Aug 2017 — May 2021**
B.S. Electrical Engineering — ***GPA: 3.47*** *Fort Collins, CO*

SKILLS & ABILITIES

Radar / RF: FMCW and pulsed radar design, pulse compression, dual-pol polarimetry, phased arrays, link budgets, Doppler processing, weather radar operations

Languages: Python (NumPy, SciPy, Pandas, PyTorch, FastAPI, Flask), Rust, Go, C / C++, SystemVerilog, Bash, SQL, Matlab, TypeScript

Design / Simulation Tools: KiCad, AWR Microwave Office, HFSS, ADS, LTSpice, Cadence, QucsStudio, Vivado

Lab / Infrastructure: VNA, spectrum analyzer, oscilloscope, noise figure and high-power test; Git, Linux, Docker, PostgreSQL, Tailscale, LaTeX, Manim

Communication: Technical presentations and design reviews (PDR / CDR), customer pre- and post-sales support, educational content reaching 50k+ engineers